

ICLR 2022 | Proving, Not Just Testing, That an RL Policy Is Robust

CROP certifies reinforcement learning policies through functional smoothing, at both the action and the cumulative-reward level

Reinforcement learning has quietly moved out of games and simulators and into places where a wrong move is expensive: lane-keeping in self-driving cars, position sizing in automated trading, control loops in robotics. In these settings an agent watches a stream of state observations and picks an action at every step. The uncomfortable part is that those observations can be tampered with. Researchers have shown that an adversary who nudges the input state by an amount too small for a human to notice can reliably flip the agent's decision, and a handful of flipped decisions is all it takes to steer a policy off a cliff.

The usual response has been to make policies empirically tougher through adversarial training, smoothness regularization, or noise injection. Each buys resistance to the attacks it was tested against, and each has, in turn, been broken by a stronger or better-adapted attacker. That cycle exposes the real gap: passing today's attacks tells you almost nothing about tomorrow's. What safety-critical reinforcement learning actually needs is a certificate, a proof that the policy stays reliable against every perturbation within a bounded budget, not merely the specific attacks that someone happened to try.

CROP, presented at ICLR 2022 by researchers from the University of Illinois at Urbana-Champaign, Washington University in St. Louis, and Carnegie Mellon University, is the first unified framework to provide exactly that for reinforcement learning. It certifies robustness at 2 levels, the stability of the action chosen at each state and a provable lower bound on the cumulative reward collected across an entire trajectory, and uses those tools to benchmark 9 existing robust RL methods across 4 environments on a common, provable footing.

Paper: <https://arxiv.org/abs/2106.09292> · Project and leaderboard: <https://crop-leaderboard.github.io>

Why robustness in RL resists certification

Certified robustness is a solved-enough problem for image classifiers, and the workhorse behind it is randomized smoothing: add Gaussian noise to the input, average the classifier's output, and the smoothed classifier comes with a provable radius around each input inside which its prediction cannot change. Lifting the same recipe straight into reinforcement learning is tempting, but two obstacles stand in the way.

The first is the output itself. In classification the confidence score lives in a known range from zero to one and behaves like a probability, which is exactly what makes the radius calculation work. A Q-network is neither: its value estimates have an unknown, unbounded range and are not probabilities at all. The second obstacle is more conceptual. Certifying a single action is not the same as certifying an agent. What ultimately matters is the reward

accumulated over a whole sequence of decisions, and a policy can be locally stable at every state yet still bleed reward under a sustained attack. CROP is built to clear both hurdles at once.

Functional smoothing: two criteria, three algorithms

The engine underneath CROP is functional smoothing. At each state, and separately for each candidate action, the method perturbs the state with Gaussian noise and averages the trained Q-network's output over that noise, producing a smoothed value function. A key lemma shows this smoothed function is Lipschitz continuous, meaning it cannot change too quickly, with a constant that shrinks as the smoothing variance σ grows. Continuity is the lever: if the value function cannot swing wildly, then a small perturbation cannot swing the chosen action either.

From that continuity the paper defines 2 certification criteria and turns each into a concrete algorithm. To keep the range problem manageable, the Q-output range is pre-estimated on a subset of states, and the smoothed values are bounded with a concentration inequality so that the certificate stays sound even under Monte Carlo sampling of the noise.

Table 1. The three algorithms in CROP, the smoothing they use, and what each one certifies.

Algorithm	Smoothing	What it certifies
CROP-LoAct	Local	Certified radius r for the per-state action; the action is provably unchanged within r
CROP-GRe	Global	Expected-reward bound J_E and the 50% percentile reward bound J_p
CROP-LoRe	Adaptive local search	Absolute lower bound J on cumulative reward, the tightest of the three

CROP-LoAct handles the action level. For each state it computes a certified radius r from the gap between the best and the runner-up smoothed action-values: as long as the perturbation is smaller than r , the top action stays on top and the policy's decision is provably unchanged. The radius exposes a genuine trade-off in σ . More smoothing makes the value function steadier, which helps, but it also compresses the margin between the top two actions, which hurts, so the best amount of noise is a balance rather than a maximum.

What the certificates reveal

Running CROP-LoAct across the 9 methods produces a per-state certified radius at every time step, which reads directly as how much perturbation each policy can absorb before its action would flip. The figure below tracks that radius over time on two Atari games, Freeway and Pong, with each column corresponding to a different smoothing level σ .

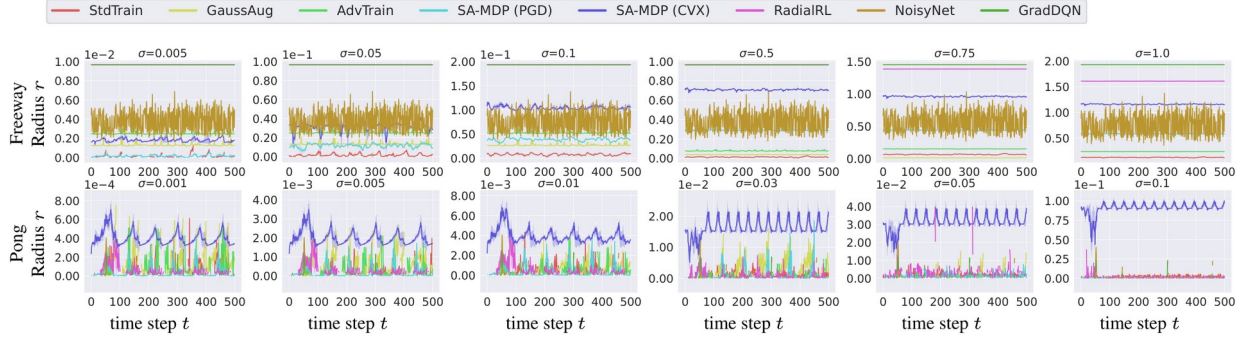


Figure 1. Per-state certified action radius r over time for all 9 methods, on Freeway (top) and Pong (bottom); each column is one smoothing variance σ . RadialRL (magenta) is the most certifiably robust method on Freeway, while SA-MDP (CVX) (purple) leads on Pong. Shaded bands are standard deviations and are small.

Two clear winners emerge, and they differ by environment. On Freeway, RadialRL holds the highest certified radius across every smoothing level, which fits its design: it explicitly optimizes against worst-case perturbations. On Pong, SA-MDP with the convex relaxation is the most certifiably robust. Reassuringly, these certified rankings line up with what people had already observed empirically. The proof agrees with the practice, and that consistency is what makes a certificate worth trusting.

The per-state view also surfaces structure that a single robustness number would hide. On Pong, every method's certified radius rises and falls in a regular rhythm rather than staying flat, and lining the curve up against the game frames explains why it does so.

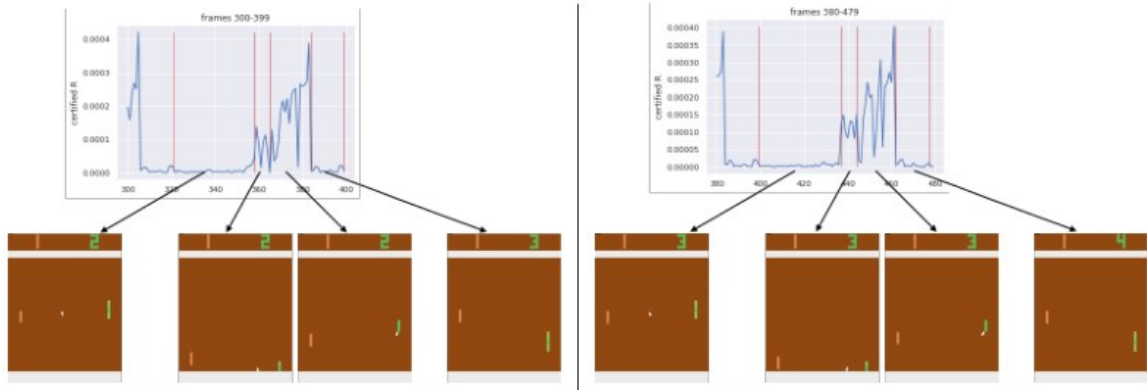


Figure 7. Two consecutive periods of the certified radius on Pong (top of each panel), aligned with the game frames they correspond to (bottom). Robustness peaks at confident moments, for example when the ball is heading toward the paddle, and dips when the next action matters less. The two periods are nearly identical.

The reading is intuitive. When the ball is racing toward the paddle the correct action is unambiguous, the margin between the best and second-best action is wide, and the certified radius is large; in the lull between rallies the decision is nearly arbitrary and the radius collapses. This suggests that robustness is not a fixed property of a policy but something that ebbs and flows with how consequential each state is, a hint that future defenses might concentrate their effort on the handful of states that actually matter most.

From single actions to whole trajectories

Certifying actions is only half of CROP. The other half asks the question a practitioner really cares about: under any bounded attack, how much reward is the policy guaranteed to collect? CROP answers with three reward bounds of increasing tightness. CROP-GRe uses global smoothing to bound the expected reward (J_E) and the 50% percentile reward (J_p), while CROP-LoRe runs an adaptive tree search over trajectories to certify an absolute lower bound (J). The figure below plots all three against the attack magnitude ϵ ; solid lines are the certified bounds, dashed lines are the empirical reward under an actual PGD attack.

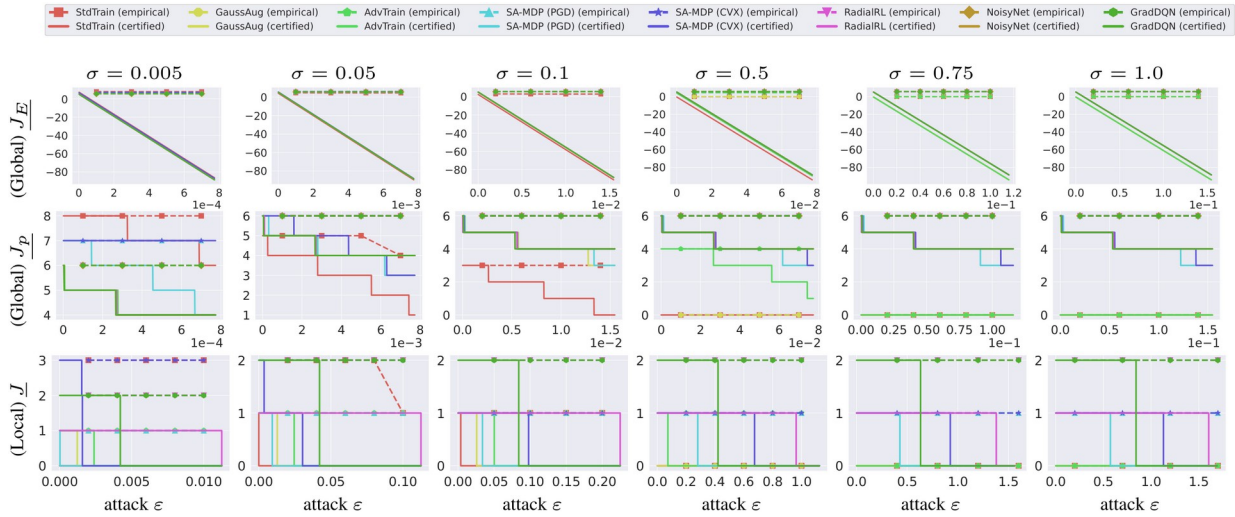


Figure 3. Certified cumulative-reward bounds versus attack magnitude ϵ for all 9 methods: the expectation bound J_E (top row), the 50% percentile bound J_p (middle), and the absolute lower bound J (bottom). Each column is one smoothing variance σ . Solid lines are certified bounds; dashed lines are the empirical reward under PGD.

The three bounds are not equally useful, and the gap between them is the point. The expectation bound J_E is loose: it drops steeply as the attack grows and quickly becomes uninformative. The percentile bound J_p is far tighter. The absolute lower bound J from CROP-LoRe is tighter still, and across a wide range of attack magnitudes it sits right on top of the empirical PGD curve, a zero gap that says the certificate is not merely valid but essentially exact. When a provable lower bound coincides with what the strongest available attack actually achieves, there is little robustness left unaccounted for.

How much to smooth

Because the smoothing variance σ controls the whole certificate, the paper studies it directly, and the answer is that the right amount of noise depends on the environment. On Freeway, which tolerates large perturbations, robustness for the strong methods keeps climbing as σ grows all the way to 1.0. On Pong the opposite holds: too much smoothing washes out the value margins, and a moderate σ between 0.01 and 0.03 is best for nearly every method. There is no universal setting. σ is a knob to tune per environment, and CROP makes the trade-off measurable instead of guessed.

Why it matters

The lasting message of CROP is that robustness in reinforcement learning does not have to be a matter of hope. By smoothing the value function you can prove that an agent's action stays fixed within a certified radius, and prove a lower bound on the reward it will collect under any attack inside a fixed budget. Applied to 9 methods across 4 environments, those certificates turn out to be not only sound but often tight, matching what real attacks achieve.

The honest boundaries are worth stating. The certificates cover L2-norm bounded perturbations of the state observation, they rest on Monte Carlo sampling with its usual statistical caveats, and they inherit the σ trade-off that has to be tuned per environment. What CROP contributes is a common, provable yardstick: instead of comparing robust RL methods by which attacks they happen to survive, the field can now compare them by what they can be proven to guarantee. The authors release their results as an open leaderboard, inviting the community to certify still more methods and more environments.